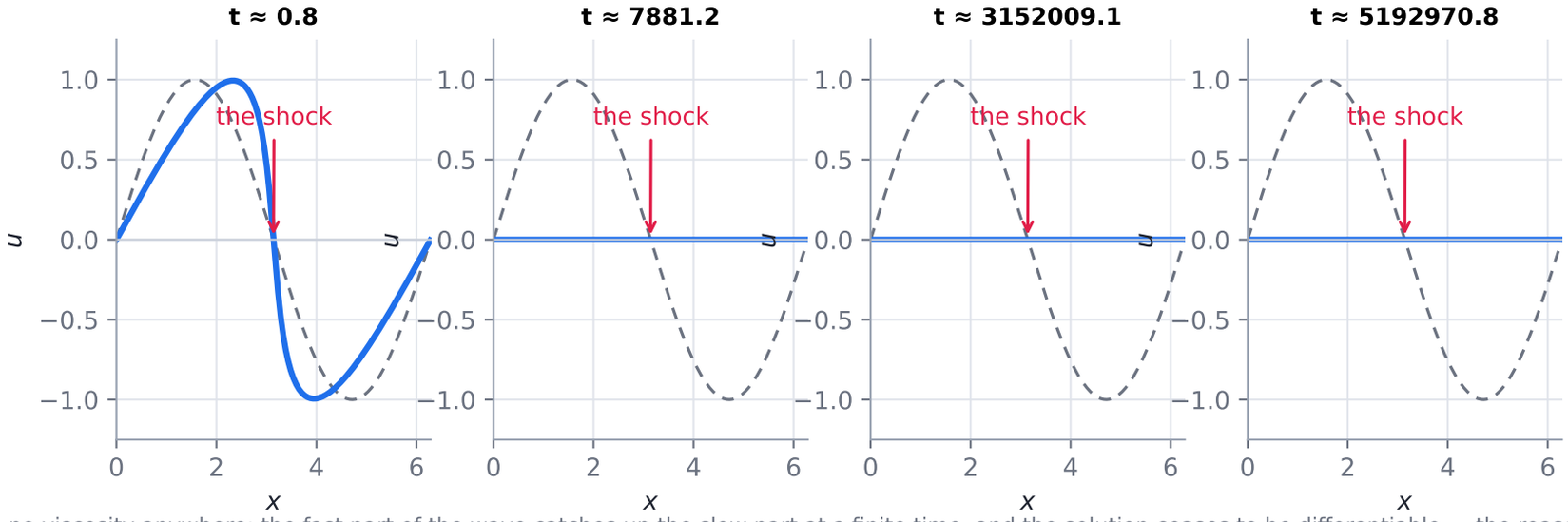

inviscid Burgers, smooth initial data, no viscosity anywhere: the fast part of the wave catches up the slow part at a finite time, and the solution ceases to be differentiable — the reason conservation laws need weak solutions.